FLEXIBARS

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AKCELPORT FLEXIBARS is formed with multiple layers of thin copper stripes, insulated with high resistance, self-extinguishing PVC compound. In this product the holes are punched directly through the laminates and are connected. The advantage of this product is that there is elimination of lugs, which eventually reduces labour and well as process cost also making the installation easier.

FLEXIBLE BUSBARS

13, Shah Industrial Estate, Deonar, Chembur, Mumbai 400088, India
Tel : +91 22 2555 5888 email : mail@akcelport.com Web : www.akcelport.com

FLEXIBARS

FEATURES :

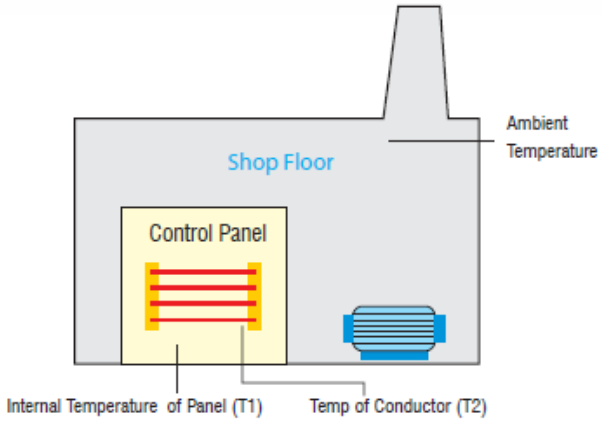
FLEXIBARS consists of copper lamination with high quality durable polymers.
Internal copper laminates are insulated in such a way that it helps to give flexibility in bending and twisting.
To get optimum surface connectivity terminals can be created by direct punch on copper lamination.
It reduces cost of labour and eliminated cost of lugs.
It is an alternative to large sizes of cables with lugs and solid rigid busbars for a panel board.
It requires less space for high ampere.
It is very easy to use as the product is flexible and easy to bend.
It is also available as per drawing.

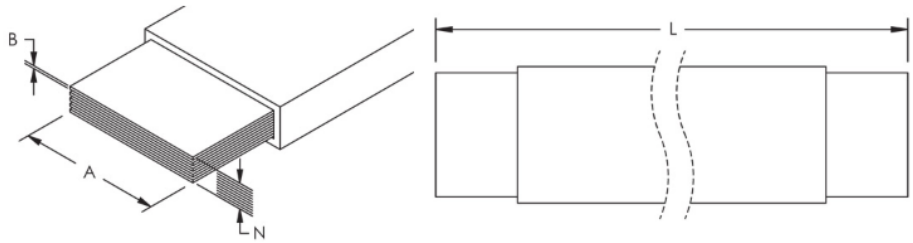
APPLICATION :

- Switchboard and Panel
- Motor Control Center
- Power Supply
- Battery Storage and Supply
- DG Set for power generation
- Control AC/ DC driven unit
- Transformer
- Electrical Machinery
- Power Convertor
- Busbar system and Link
- UPS system
- Windmill Power generation unit
- Combination motor starter, controller
- Railways, coach & power supply system
- Requirement of high ampere with small conductor size

SELECTION OF FLEXIBARS :

Ex : For a current of 400A with
T1 = 40° C - T2 = 90° C
1) Ä T = 90 - 40 = 50° k (T2- T1)
2) In 50° C column, find the closest current value to 400A
3) Size : 5 x 20 x 1.00mm
OR 3 x 24 x 1.00mm = 413 Amp.
Select according to size of Terminal.





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FLEXIBARS SIZES & AMPLICITY TABLE:

A	Part Code			ΔT(K)					Current Coefficient		Length 2M
		N A B	sqmm	70	60	50	40	30	2	* 3	*Pack qty
125A	1258605	8 X 6 X 0.5	24	196	182	166	143	128	1.72	2.25	10
	1253908	3 X 9 X 0.8	21.60	158	147	134	120	104	1.72	2.25	10
	1253135	3 X 13 X 0.5	19.50	198	184	167	150	130	1.72	2.25	10
	1252158	2 X 15.5 X 0.8	25.80	252	234	212	191	165	1.72	2.25	10
250A	2504158	4 X 15.5 X 0.8	49.60	380	350	320	286	248	1.72	2.25	10
	2502201	2 X 20 X 1	40	326	300	275	246	214	1.72	2.25	5
	2503201	3 X 20 X 1	60	428	395	360	323	280	1.72	2.25	5
	2502241	2 X 24 X 1	48	450	416	380	340	295	1.72	2.25	5
	2506135	6 X 13 X 0.5	39	300	277	253	226	196	1.72	2.25	5
400A	4005201	5 X 20 X 1	100	498	460	420	376	326	1.72	2.25	5
	4003241	3 X 24 X 1	72	490	453	413	370	320	1.72	2.25	5
	4004241	4 X 24 X 1	96	550	510	465	416	360	1.72	2.25	5
	4002321	2 X 32 X 1	64	480	445	406	363	315	1.72	2.25	5
	4002401	2 X 40 X 1	80	538	500	455	406	352	1.72	2.25	5
	4006158	6 X 15.5 X 0.8	74.4	476	440	402	360	318	1.72	2.25	5
500A	5005241	5 X 24 X 1	120	608	563	514	460	398	1.72	2.25	5
	5004321	4 X 32 X 1	128	648	600	548	490	425	1.72	2.25	5
630A	6308241	8 X 24 X 1	192	802	743	678	606	525	1.72	2.25	5
	6305321	5 X 32 X 1	160	758	702	640	573	496	1.72	2.25	5
	6306321	6 X 32 X 1	192	846	783	715	640	555	1.72	2.25	5
	6305401	5 X 40 X 1	200	900	832	760	680	590	1.72	2.25	5
	6301201	10 X 20 X 1	200	762	706	645	576	500	1.72	2.25	5
800A	8000241	10 X 24 X 1	240	948	877	800	716	592	1.72	2.25	5
	8008321	8 X 32 X 1	256	1018	943	860	770	667	1.72	2.25	5
	8006401	6 X 40 X 1	240	1018	943	860	770	667	1.72	2.25	5
	8005501	5 X 50 X 1	250	1100	1016	930	830	718	1.72	2.25	5
1000A	1008401	8 X 40 X 1	320	1230	1140	1040	930	805	1.72	2.25	3
	1008501	8 X 50 X 1	400	1393	1290	1175	1050	912	1.72	2.25	3
1250A	1251501	10 X 50 X 1	500	1650	1525	1395	1245	1080	1.72	2.25	2

Note:

This table indicates the temperature rise produced by chosen current in the given section.
This calculation does not take into account the heat dissipation from the switchgear.
Any other size on request with Minimum Order Quantity
Available in 2 & 3 meters of length in red copper.

* When using 2 or 3 FLEXIBARS in parallel use the current coefficient at 50 C (T)

Example: 640 Amps

Amps capacity = 2 bar Parallel 640A x 1.72 = 1100A

= 3 bar Parallel 640A x 2.25 = 1440A

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TECHNICAL CHARACTERISTICS :

- Weight and volume saving within the panel, i.e. cross section area 40% lower than a cable.
- Allows ultimate twisting and bending
- Time saver while assembling or installing
- Easy to use and cost effective
- Aesthetically improves the appearance of the panel.
- Improves safety and reliability of your panel due to the reduced amount of connection
- Gauranteed isolation thickness, including the corner

CONDUCTOR :

- Copper Strip Electrolytic grade (>99.9% Cu)
- Surface : Uncoated bare
- Tensile : > 200 N/mm
- Extensibility : > 20% (minimum)
- Lamination Thickness : 0.5mm to 1 mm

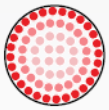
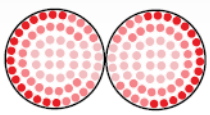
INSULATION :

- High Grade polymer
- Color : Black
- Thickness : 2mm +/- 0.2 mm
- Shore Hardness : A 85 (I3)
- RoHS Compliance
- Electrical Strength : > 20 Kv/mm
- Flammability : UL 94 VO
- Tensile Strength : > 9 N/mm
- Elongation : > 250%
- Dielectric Rigidity : 20 Kv/mm

FINISHED PRODUCT :

Operation temperatures : -20 deg Celsius to +105 deg Celsius
Operating Voltage : 1100 Volts AC & 1500 Volts DC
Test Voltage : 3000 Volts/5 minutes

SKIN EFFECT :

Comparison of the penetration depth between Copper CABLE & FLEXIBAR		
250 A	 1 X 95 SQMM COPPER CABLES 95 SQMM	 1 x FLEXIBAR 2 x 20 x 1 40 sq mm
630 A	 2 X 150 SQMM COPPER CABLES 300 SQMM	 1 x FLEXIBAR 5 x 32 x 1 160 sq mm

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CUSTOM MADE FLEXIBLE BUSBARS:

Cutting | Bending | Twisting | Hole pattern | Oblong holes



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